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SECTION A - DATA FUNDAMENTALS

① A Database - is an organized collection of information stored in tables so that it can be easily accessed, updated, searched, and managed.

* Why are they important - They help store information in an organization safely and use the information effectively and in real time.

* Real life examples

- Banks use Databases to store customer information bank balances, process ~~trans~~ transactions. When transactions are made, the database is updated immediately.

- Hospitals use databases to store patient information medical records, test results, and billing. This helps the hospital to easily access the information and to provide appropriate care.

② A Data Warehouse - It is a large centralised system used to collect and store data from different sources/systems so that the organization can analyse the data, identify trends to make better business decisions. E.g. A retail company can would combine sales, customers, inventory, and online shopping in a data warehouse. It uses ETL to Clean, Prepare, and organise data for Business Intelligence.

* The difference -

Databases

- mainly store day to day transactions
- Data is frequently updated
- Designed for fast individual transactions

Data Warehouse

- Mainly used for Analysis and reporting
- Data is updated at the end of day
- Designed for complex queries and large scale analysis

(2)

* Companies use Data Warehouse to bring information from different systems into one place. to

(3) A data lake is a repository designed to store large volumes of raw data, using low-cost ~~data~~ cloud object storage. It allows the ingestion and storage of structured, semi-structured, and unstructured data within a single platform.

* Differences

Data lake

- Stores data in its raw format
- Data can be stored without knowing how it will be used
- Flexible for exploration

Data Warehouse

- Stores cleaned and structured data
- Data is usually structured according to specific business and reporting requirements
- Optimized for fast reporting and Business Intelligence.

A business would prefer a data lake when it has large amounts of different data types. The business might opt for data lake when they have data that they do not know how to use it yet.

(4) Data Mart - It is a storage system that contains information specific to an organization's department. It contains a small and selected part of the data that the company stores in a larger storage system. Companies use a data mart to analyze department-specific information more efficiently.

(3)

* The difference

Data Mart

Stores a smaller subset of
of a bigger ^{data} integration
of the whole organization.
- Business Unit focused

Data Warehouse

- stores and integrates data
from across the organization
- Focuses on the entire organization
across.

* Who uses it - Data marts are typically used by specific departments and their analysts or managers.

example: Sales team to analyse sales performance and revenue
: HR teams to analyse employee information and work trends.

⑤. - Structured data. This is data that is highly organized and follows a predefined format or schema. It usually fits neatly into rows and columns.

Semi-Structured

Example: Spreadsheets.

- Semi-Structured data - Does not follow rigid table format, but it contains tags, metadata, labels or other markers that give the data some organization. It sits between structured and unstructured data.

Example: JSON

```
{ "name": "terato",  
  "Age": 25,  
  "City": "Pretoria" }
```

The tags "name", "age", "city", tell the computer what each piece of information represents.

- Unstructured data - It does not have a predefined structure that fits into rows and columns, ...

④
... Unstructured data continued from page 3.

Unstructured data can include text, images, audio, video, and other forms of information.

Example: A customer sends a company a voice note/recording explaining a complaint.

⑥ Data Modelling - It is a key process in designing and organizing data structures to support efficient storage, retrieval and ~~system~~ analysis of information. It is the architectural foundation for any data warehousing system. Effective data modelling can help organizations unlock the full potential of their data by analyzing and defining the different types of data an organization collects and illustrating the connections between those data points and structures.

It is a structured representation of various texts, symbols, and diagrams that demonstrates how data is stored, organised, and accessed, making it easier to design and manage databases effectively.

* Types of ~~the~~ data models

① Relational data model - It organises data information into tables consisting of rows and columns. Tables are connected using relationships, usually through primary and foreign keys. Relational Model is widely used in relational databases management systems such as MySQL.

② Dimensional data Model - It is commonly used for data warehouses, business intelligence, reporting and analytics. It organises data into fact tables and dimension tables.

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4. ~~Data~~ Modelling

8. Database Management System (DBMS) - It is a software that allows users and organizations to create, store, organize, manage, retrieve, update, and protect data in a database.

* Four popular DBMS's :

① MySQL - An open source DBMS that uses SQL. It is widely used for web applications and business systems.

Use Case - E-commerce website: managing customers, products, orders, payments and inventory.

② Oracle database - It is an enterprise level DBMS designed for organizations that require high performance, security, scalability, and reliability.

Use Case : A bank could use Oracle Database to manage customer accounts and millions of daily financial transactions. The database needs to ensure that the transactions are processed accurately and that customer information is protected.

③ Microsoft SQL Server - It is a relational DBMS developed by Microsoft. It is commonly used by organizations to manage operational and analytical data.

use Case : A retail company could use SQL server to store sales and customer information and then connect the data to PowerBI to create dashboards showing sales by product, region, and month.

* PostgreSQL is an open source object relational database system. It is particularly suitable for applications requiring complex queries, strong data integrity, and advanced data type.

Use Case : A research organisation could use it to store and analyse large data sets from different research projects.

⑦

9. ETL (Extract Transform Load). It is a process used to collect data from different sources, prepare and clean that data, then store the data in a destination such as a data warehouse so that it can be analyzed.

ETL is important in data analytics because data comes from different ~~sources~~ systems and may be incomplete, duplicate, inconsistent, or stored in different formats.

* The ETL Steps

- ① Extract - Collect data from different sources without changing its original source unnecessarily.
- ② Transform - This step involves cleaning, changing, and preparing the extracted data for meaningful and reliable analysis.
- ③ Load - Moving the transformed data into its final destination.

10. Common data storage formats in data analytics - They are formats in which data is structured and saved so that it can be stored, processed, and analysed. The best format depends on data type, size of the dataset, how frequently it changes, and what tools will be used to analyze it.

Format	Description	When to use it.
• CSV	Plain text file where data is separated by ; delimiter.	Small to medium tabular datasets, data exchange, and simple analysis in excel, python or SQL.
• Excel	Spreadsheet format can contain multiple worksheets, formulas, formatting, tables and charts	Business users working with relatively small data sets and doing manual analysis and reporting
• Parquet	A Column oriented, compressed file format designed to store and analyse large data sets	Big data analytics, data lakes spark/python or workflows.

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Format	Description	When to use it
• Avro	A compact, row-oriented, data serialization compressed file format that includes a Schema	Data pipelines and streaming systems where data is frequently written/read as complete records and schema need to be managed.
• JSON	Stores data as key value pairs and nested objects. It is commonly used for web applications and APIs	Web/API data, application data, and data sets with nested or flexible structures.

1. Section B: Artificial Intelligence and Machine learning.

11. Artificial Intelligence - Is the ability of a computer system or machine to perform tasks normally require human intelligence. These tasks can include learning from data, recognize patterns, understanding language, solving problems, Making predictions, and making decisions.

- Narrow AI (Weak AI) - It is an AI system designed to perform specific task or limited range of tasks. It does not have general human-like intelligence and cannot automatically apply it's abilities to every type of problem.
- General AI (Strong AI) - It refers to a hypothetical AI system that would have the ability to understand, learn, reason, and perform a wide variety of intellectual tasks at a level comparable to humans.

Examples :

Weak AI	Strong AI
Designed for • Voice assistants • Spam filters	• Write computer programs

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12. Machine learning (ML) - It is a branch of artificial intelligence that enables computers to learn patterns from data and use those patterns to make predictions or decisions without being explicitly programmed for every individual situation.

The difference - The main difference is how the computer learns to produce an answer.

• Traditional programming:

Rules / Instruction + Data = Output.

A programmer creates specific rules for the computer to follow.

• Machine learning: Data + Expected results $\rightarrow$ ML models $\rightarrow$ predictions.

Instead of writing every rule manually, the computer is given data and an appropriate ^{learning} algorithm. It identifies patterns in the data and builds a model that can be used to make predictions on new data.

Three main types of ML

• Supervised learning - Uses training data that has known labelled outcomes

• Unsupervised learning - Uses data that does not have predefined labels or answers. The algorithm attempts to discover patterns, structures, or groups within the data.

• Reinforcement learning - It involves AI agent learning by interacting with an environment. The agent receives rewards or penalties based on its actions and gradually learns which actions produce better results.

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13. Deep learning - It is a specialized branch of ML that uses artificial neural networks with multiple layers to learn complex patterns from large amounts of data.
* How does deep learning relate to AI and ML - All deep learning is machine learning, and machine learning is a type of AI, but not all AI is ML and not all ML is deep learning.

* Neural Network is a machine learning model inspired loosely by the way biological neurons or nodes are arranged into layers.

A basic neural network has three types of layers.

- ① Input layer - Receives data
- ② Hidden layer - Processes the information and identifies paths.
- ③ Output layer - Produces the final predictions or classification.

14. What is Natural Language Processing (NLP) - It is an AI branch that enables computers to understand, interpret, process, and generate human language. Unlike ^{complicated} human language, NLP uses techniques from ML, linguistics, and computer science to help computers work with text and speech.

* 3 real-world NLPs:

- ① Voice assistants
- ② Chatbots and customer services
- ③ Predictive texts and auto correct.

15. What is large language model (LLM) - It is a type of AI model trained on very large amounts of text so that it can understand and generate human language. They are generally based on deep learning ~~and~~ neural networks particularly the transformer architecture.

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... How is LLM different from a standard LM model
The main difference is on their purpose, scale, and flexibility.

LLM	Standard LM model
- Designed to understand and generate language. - Trained on large amounts of data. - Can perform many different language tasks.	- Designed to solve a specific prediction or classification problem. - Often trained on a data set specific to one task. - Usually performs narrow tasks.

* Two well known LLMs.

① GPT (Generative pre-trained Transformer) is a family of large language models developed by open AI. They can be used for conversational AI, Summarisation, coding assistance e.t.c

② Google Gemini - It is a family of AI models designed to work with different types of information, including text, images, audio, video, and code, depending on the model and application. It can be used for: Answering of questions coding e.t.c.

16. Generative AI - It is a type of AI that can create new content based on patterns it has learned from existing data.

- How does it work: At high level generative AI works through several stages.

1. Training - trained on existing data.
2. Learning patterns: The model sees and stores copies of everything.

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... ④ Generation - The model uses what it learned during training to generate new content that matches the ~~new~~ user's request.

⑤ Output - The system produces the requested content, such as advertisement, Image or other output

* Three generative AI tools:

① Chat GPT - It is developed by open AI. It can generate and work with text.

② Google Gemini - It is from Google, ^{it} can understand and generate content across modalities, depending on the specific gemini model and product.

③ Microsoft Copilot - It is from Microsoft. It can help users work and create content. It is integrated into ~~several~~ various Microsoft products and services, depending on the version and license.

17. Training Dataset - Is a collection of data that is used to teach an AI machine learning model how to recognise patterns, make predictions, classify information, or generate outputs.

* The importance of data quality when training - It directly affects the accuracy, reliability, and fairness of the AI model. AI models learn patterns from the data they are given, so poor-quality data can cause the model to learn incorrect patterns. Inaccurate predictions, biased results.

18. Computer Vision - It is a branch of AI that enables computers and machines to process, analyse, and interpret visual information. It allows the machine to identify the objects, people, text, patterns, movements, and other ~~visually~~ features in visual data.

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... Machines do not 'See' images in the same way humans do. Computer represents an image as pixels. With each pixel containing numerical information about characteristics such as colour and brightness.

* Two real-world use cases:

- ① Health Care - Medical image analysis.
- ② Manufacturing - Quality control ~~technology~~.
Inspection of defective products.

14. SECTION C: API, TOOLS, AND DEVELOPER CONCEPTS

19. API - It stands for Application Programming ^{Interface} ~~Language~~.

It is a set of rules and methods that allows different software applications or systems to communicate with each other and exchange information.

* Analogy: An API is like a waiter in a restaurant. A waiter serves as a middle person between the client and the kitchen.

* APIs uses - to ^{enable} communication between applications.
- Connect different software applications/systems.

* The importance of APIs - They allow different systems to communicate

- They make data accessible
- They support automation
- They make software development effective.
- They improve ~~cloud~~ integration between cloud services.

20. API Keys - It is a unique string of characters used to identify and authenticate an application or user when making request to an API.

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••• Purpose of an API - Authentication and Identification

- Controlling API usage
- Monitoring usage
- Access control

* Storage and protection of an API key - It should be treated as sensitive information, particularly if it provides access to paid services or private data.

- Do not put it in code.
- ~~Put it~~ Store it outside the source code.
- Restrict the API key.

* What could happen if the API key is exposed - This will lead to unauthorised usage

- Unexpected costs
- Access to protected resources
- Reputation and security problems.

21. Rest API - It stands for Representational State Transfer Application programming interface. Rest API allows different applications to communicate with each other over the Internet using the HTTP protocol. It uses methods to connect with different resources.

* Main Rest API methods

~~Rest~~

METHOD	PURPOSE	Simple Example
GET	Retrieve data	Get a list of products
POST	Creates new data	Add a new product
PUT	Updates/replace an existing resource	Replace a customer's details
PATCH	Partially updates a resource	Change only a customer's ^{information} product details . (some information).
DELETE	Removes data	Delete a product

⑮ JSON (JavaScript Object Notation). It is a lightweight-text based format used to store and exchange data between computer systems. Although JSON originated from Java Script, it is language-independent, meaning it can be used with many programming languages, including Python, Java, C++
 JSON is common in REST APIs and information exchanges because it is simple, compact and easy for both computers and humans to understand. It is easy to read, It is easy for computers to process, It works across different programming languages, It supports different types of data.

JSON Object:

```
{
  "customer_Id": 1001,
  "Name": "Sarah M",
  "email": "email@email.com",
  "age": 25,
  "active": true
}
```

Key	Value	Data type
customer_Id	1001	Number
Name	Sarah M	String
email	email@email.com	String
age	25	Number
active	true	Boolean Boolean

②③ Prompt Engineering - Is the process of designing and refining instructions or questions (prompts) given to an AI system so that it produces a more useful, accurate and relevant response.

* Why is it an important skill - It helps produce more relevant answers.

- It improves the quality of AI-generated content

- It can save time

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• It is useful across many AI applications

* Poorly written prompts -

* Explain machine learning

Problem - This is broad. It does not tell the AI:

- Who ^{the} explanation is for
- How detailed it should be
- What aspects should be covered
- What format is preferred

Improved Prompt - Explain machine learning to a beginner who has no programming experience. Define machine learning in simple terms, explain the 3 main types of machine learning, and give one real-world example of each. Use headings and a simple comparison table. Keep the explanation under 500 words.

* Analyse this sales data

Problem - The AI doesn't know what kind of ~~data~~ sales analysis you want or what business question you are trying to answer

Improved Prompt: Analyse the attached sales dataset for a retail business. Identify the three products with the highest sales, calculate the total revenue by month, identify any noticeable sales trends, and provide three business recommendations based on your findings. Present the results in a clear table and explain the findings in simple language for a business manager

24) MAP Function - Is a programming function used to apply the same operation or function to every item in a collection of data, such as a list or array

⑦ A MAP in programming generally:

- Takes a collection of values
- Goes through each value
- Applies a function to that value
- Collects the results
- Returns the transformed collection.

Python example

</> Python

```
numbers = [1, 2, 3, 4, 5]
results = map(lambda x: x*2, numbers)
Print(list(results))
```

Output

[2, 4, 6, 8, 10]

The function $\text{Lambda } x: x*2$, takes each value x and multiply by 2.

* MAP in Data processing

</> Python

```
Prices = [100, 200, 300, 400]
Add 15% Vat
```

```
Prices = [100, 200, 300, 400]
```

```
Prices_with_vat = list(map(lambda price: price*1.15, Prices))
Print(prices_with_vat)
```

Output

[115.0, 230.0, 345.0, 460.0]

* This is useful in data analytics:

- for converting currencies
- Cleaning text
- Converting data types
- Calculating new values

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26. Cloud Computing - Cloud Computing is the delivery of computing resources and services such as servers, storage, databases, networking, software and processing power over the internet, rather than requires users or organisations to own and maintain all the physical infrastructure themselves.

* The three main cloud Service Models

1. IaaS - Infrastructure as a service. Provides basic computing infrastructure over the internet, including things such as: Virtual machines/servers

Example: ~~Amazon~~ ^{Amazon} EC2

Storage

Networking

Operating systems environments

2. PaaS - Platform as a service. PaaS provides developers with a ready-made environment for developing, testing, deploying, and managing applications.

Example: Google App Engine [from google cloud].

3. SaaS - software as a service. The cloud provider delivers a complete software application over the internet. Users don't generally need to install or maintain the underlying infrastructure or application software themselves.

Example: Microsoft 365 - Users can access applications such as Word, Excel e.t.c. through the internet without having to manage the OS, Servers, or underlying infrastructure.

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27. Jupyter Notebook - is a free, open source interactive ~~environ~~ computing environment that allows you to write and run code, view the results, create visualisations and document your work in the same place. It is made up of cells. You can put Python code in one cell, run it, and immediately see the output, and then continue with the next step. You can add ^{exploratory} ~~exploratory~~ text, equations, images, and charts. It is an application for creating and sharing computational documents containing code, narrative text, ^{equations,} and rich outputs.

Jupyter and programming languages - It is commonly associated with Python particularly in Data Analytics, Data science, and machine learning. It is however not limited to Python. It supports more than 40 programming languages including R, Julia, and Scala. It uses a kernel (component) to run code in particular programming language. e.g. Python kernel to run Python code and send results to the notebook.

Why is it popular with DA and DS?

1. It allows interactive data analysis
2. It works well with Data Science libraries.
3. It combines code, results, and explanations
4. It is excellent for data visualisation.
5. It supports experimentation

28. Python - It is a high-level, general-purpose programming language known for its simple, readable syntax and broad range of applications. It can be used for web development, automation, software development, data analysis, ML, and AI.

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- ... Why is it popular for DA and AI - Python is relatively easy to learn than other programming languages. This makes it practical for people entering data analytics, data science and AI.
- ② It has a large ecosystem of data and AI libraries
 - ③ It works well with Jupyter Notebook.
 - ④ It is widely used in AI and ML.
 - ⑤ It has a large developer community.

Four Python Libraries

- ① Pandas - It is used for data manipulation and analysis. Useful for working with tables of data, such as CSV and Excel files. [Load datasets, clean data, filtering, grouping, calc, combining DS].
- ② Numpy - It is designed for numerical and scientific computing. It provides sufficient arrays and mathematical operations that are widely used as a foundation for data science and machine learning. It is useful when working with large amounts of numerical data and mathematical calculations.
- ③ Matplotlib - It is used to create visualisations and charts. You can create Line, Bar, Histogram, Scatter plots, and Pie charts. A data analyst can use it to turn numerical results into visual information that is easier to understand.
- ④ Scikit-learn - It is focused on machine learning. It provides tools for tasks such as:
Classification, regression, clustering, predictive modelling, model evaluation, Data Preprocessing.

DA.

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Data Ethics, Governance, and Careers

21. Data Privacy - Refers to the proper handling of personal information. How data is collected, used, stored, shared, protected, and deleted forms part of data Privacy. Individuals have rights over the information mentioned above.

The importance of data Privacy - It is important because personal information can be valuable and, if misused or exposed, can cause significant harm.

- ① It protects individuals
- ② Builds trust.
- ③ Reduces security risks
- ④ Helps organisations to comply
- ⑤ Gives individuals control over their information.

What is GDPR (General Data Protection Regulation)

~~and data~~ It is regulation (EU) 2016/679, a comprehensive data-protection law that regulates the processing of personal data and protects the rights of individuals. It gives individuals stronger rights over their personal information and establishes common data-protection rules across the EU.

Countries that are covered by the GDPR

Not only EU countries are subjected to the GDPR, even countries outside EU are. All countries that

- ① Offer goods and services to individuals in the EU
 - ② Monitor behaviour of individuals in the EU
- } GDPR
APPLIES

22. Data Ethics - It is the study and application of moral principles to how data is collected, stored, analysed, and shared / used. It focuses on ensuring that data is used in ways that are fair, responsible, transparent, respectful of privacy and beneficial to people. ...

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... Data ethics is closely connected to AI ethics, because AI systems learn from and make decisions using data. Ethical AI therefore requires attention to issues such as privacy, fairness, transparency, accountability, and harmful bias.

Examples of Unethical Use of Data and AI

① Using biased data to make decisions about people.
This is problematic because AI may ^{reproduce} use or even amplify existing discrimination.

② Using or collecting data without proper consent or transparency.

This is problematic because it can expose people to identity theft, profiling, manipulation or unwanted surveillance.

③ Using AI to manipulate or deceive people

This is problematic because the technology is used to deceive instead of providing useful information. This may lead to people making decisions using false info.

31. Data Governance - Is the system of policies, rules, processes, roles, and responsibilities that an organisation uses to ensure that its data is accurate, secure, available, properly managed, and used responsibly.

Key components of good data governance framework

Data governance framework is essentially an organisation's blue print for managing its data. There is no single framework that each organisation must use. The adaption will be from the organization's size, industry, regulations, technology, and business objectives

① Data governance and strategy

② Roles and responsibilities

③ Data policies, standards, and procedures

④ Data quality

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- ... ② Data Security and access control
- ③ Data privacy and compliance
- ④ Data lineage
- ⑤ ~~Customer database~~ Data Classification.
- ⑥ Monitoring, Auditing, and continuous improvement
- ⑦ Meta data and data cataloging.

32. Bias in AI occurs when artificial intelligence system produces systematically unfair, inaccurate or discriminatory results, often because of biases in the data, the way the model was designed, or how the system is used. An AI model is considered biased when its predictions or decisions consistently disadvantage certain people or groups, rather than treating comparable cases fairly.

A real world example: Amazon's recruitment AI.

Amazon developed an AI-based tool intended to help evaluate job applications. The system was trained by approximately 10 years of historical résumé data. Because the historical application pool was predominantly male, the training data reflected an existing gender imbalance. The system consequently learned patterns that tended to favor male applicants.

Reports indicate that the tool penalised some resumes containing wording associated with women and downgraded graduates of women colleges.

33. Data Analyst Vs Data Scientist Vs Data Engineer Vs AI Engineer

Role	Main focus	Skills	Tools
① Data Analyst	Turning data into meaningful insights	SQL, Excel, basic stats	SQL, Excel, Power BI, Tableau, Python
② Data Scientist	Explore data, build Predictive models, Perform Statistical analysis, develop and evaluate ML models	Python/R, Statistics, ML Probability, SQL, maths	Python, Pandas, Numpy, Scikit-learn, Jupyter, Tensorflow, Pytorch, SQL
③ Data Engineer	Building and maintaining data infrastructure and pipelines	Collected data, build ETL/ET, Cloud computing, data architecture	Python, SQL, Apache, Spark, Airflow, dbt, AWS, Azure, GCP, Snowflake, databricks
④ AI Engineer	Building and deploying AI/ML systems into real applications	Develop, Integrate, deploy, and monitor AI Models. Build AI Powered applications, optimise Models for Production	Python, Software Engineering, ML/deep learning, API, Cloud, Mlops, Model deployment

34. Business Intelligence - It refers to the technologies, processes, and practices that organisations use to collect, analyse, and present data so that people can make better business decisions.

Popular BI Tools

① Power BI is a business Analytics platform that allows users to connect to data sources, transform data, and analyse data, and create interactive reports and dashboards.

② Tableau is a data visualisation platform designed to help people explore data, identify patterns and communicate insights through interactive visualisations and dashboards.

SECTION E: PERSONAL REFLECTION

35. ~~W3S~~ Data Fundamentals. I think ^{the} overall ~~the~~ depth and architected involving data and the beauty of how data works, what is involved in sourcing it. The most important thing I have learnt is the different ways data can be stored and managed. I have realised that I had limited knowledge to the technologies and tools available out there that one can use in the Data world.

36. Python Map function was a bit difficult for me to understand because I have never heard of it before. I read + practised more from W3Schools and Youtube.

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37. How data and AI will affect my future career and everyday life.

I believe data and AI will have a significant impact on my future career and everyday life. As I develop my skills in data analytics, I expect data and AI to become tools that help me work more efficiently, make better decisions, and solve problems.

I could use SQL to retrieve information from a company's DB, Python and Pandas to clean and analyse the data, the Power BI to create dashboards showing important business trends.

In my everyday life, I already see how data and AI can make activities more convenient. E.g. For example, online shopping platforms use data about my previous searches and purchases to recommend products.

* Please see next page for References.